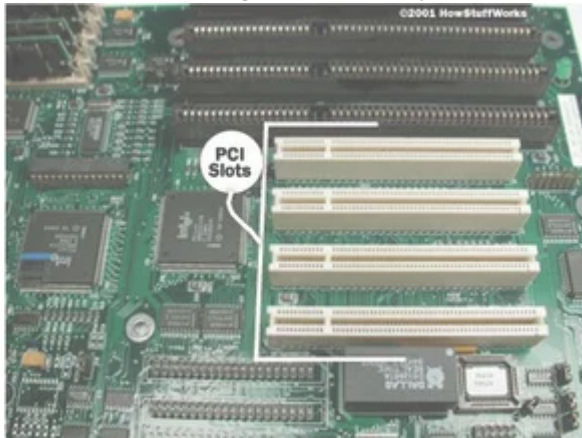


Configuring PCI Cards

Peripheral Component Interconnect
used for connecting internal components



- the PCI bus is the standard expansion bus for internal devices
- plug-and-play style configuration.

Configuration

most PCI devices configure themselves automatically.

however you can tweak how PCI devices are detected:

- The kernel has options that affect how it detects PCI devices. These are in the kernel config screens under Bus Options.
- Most firmware has PCI options that change the way PCI resources are allocated.

setpci

query and adjust PCI devices configurations

lspci

- info about the PCI busses on your system
- items connected to those busses

options

`-v, -vv, -vvv` verbose, (-vv and -vvv increase verbosity)

```
james@laptop:~$ lspci -vt
-[0000:00]--+-00.0 Intel Corporation 8th Gen Core Processor Host Bridge/DRAM Registers
+-01.0-[01]--+-00.0 NVIDIA Corporation GP107GLM [Quadro P1000 Mobile]
|               \-00.1 NVIDIA Corporation GP107GL High Definition Audio Controller
+-02.0 Intel Corporation CoffeeLake-H GT2 [UHD Graphics 630]
+-04.0 Intel Corporation Xeon E3-1200 v5/E3-1500 v5/6th Gen Core Processor Thermal Subsystem
+-08.0 Intel Corporation Xeon E3-1200 v5/v6 / E3-1500 v5 / 6th/7th/8th Gen Core Processor Gaussian Mixture Model
+-12.0 Intel Corporation Cannon Lake PCH Thermal Controller
+-14.0 Intel Corporation Cannon Lake PCH USB 3.1 xHCI Host Controller
+-14.2 Intel Corporation Cannon Lake PCH Shared SRAM
```

Kernel Modules

- **provide access to hardware**
- **a driver is a type of module but not all modules are drivers**
- the stand-alone files are stored in *lib/modules*

Linux loads the modules when it boots, but you can load additional modules.

lsmod

shows all the modules that are loaded

```
james@laptop:~$ lsmod | grep -E ' 1 |Used by'
Module                Size  Used by
snd_hda_codec_generic 122880  1 snd_hda_codec_realtek
snd_hda_scodec_component 20480  1 snd_hda_codec_realtek
platform_profile      12288  1 dell_pc
cdc_ether              24576  1 r8153_ecm
intel_uncore_frequency_common 16384  1 intel_uncore_frequency
snd_sof_intel_hda_generic 36864  1 snd_sof_pci_intel_cnl
```

Used by - number of dependencies (other modules)

if it's being used by another module, the module name is shown next to the number

Loading Kernel Modules

two programs to load them: `insmod` `modprobe`

Linux loads modules automatically

insmod [path]

must **manually** load module dependencies

you can pass in module options, that are specified in a module

example

```
# insmod /lib/modules/3.16.6/kernel/drivers/bluetooth/bluetooth.ko
```

modprobe [name]

automatically loads dependencies

```
# modprobe bluetooth
```

options	
<code>-C filepath</code>	modprobe configuration file is /etc/modprobe.conf or in /etc/modprobe.d/ you can change it by passing in your own filepath and file
<code>-r --remove</code>	remove module and any on which it depends that are not used elsewhere
<code>--show-depends</code>	show modules that the specified module depends on

Removing Kernel Modules

Modules consume a small amount of memory

rmmod

```
rmmod modulename
```

options	
<code>-f --force</code>	force
<code>w --wait</code>	wait for module to become unused then unload it